3-Week Intensive Syllabus

Programme structure and assessment

SpaceCDF -- AI-Assisted Concurrent Design Facility
University of Ottawa - SEDTI
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DRAFT

Space Mission Design & Operations Programme

3-Week Intensive (Monday-Friday, 9:00-16:00)

Audience: Industry professionals, graduate students, upper-year undergraduates

Capacity: 20-30 participants in teams of 4-5

Tool: SpaceCDF AI-Assisted Concurrent Design Facility

WEEK 1: Canadian Space Landscape & Regulatory Environment

Day	Topic	Activities
Mon	The Canadian Space	Canadian Space Agency mandate & programs. Canadian space industry map (MDA, Telesat, GHGSat, Kepler, NorthStar, university labs). CSA funding mechanisms (STDP, FAST, CSEP). Canada's role in international partnerships (ISS, Lunar Gateway, Radarsat). Guest speaker: CSA or industry representative.
Tue	Regulatory Framework	ISED spectrum licensing (CPC-2-6-02). RSSSA (Remote Sensing Space Systems Act) for EO missions. Canadian Controlled Goods Program and export control. ITU filing process through ISED. Practical: complete an ISED spectrum licence application for a sample mission.
Wed	International Standards &	ECSS standard framework and tailoring for small missions. NASA SEH 17 processes overview. CubeSat Design Specification (Cal Poly CDS Rev 14). Space debris mitigation (IADC 25yr, FCC 5yr). UN Registration Convention (COPUOS). Practical: complete a tailoring matrix.
Thu	Mission Needs & Opportuniti	Identifying mission needs in the Canadian context (Arctic monitoring, maritime surveillance, agriculture, resource management, connectivity). Stakeholder analysis for Canadian missions. Indigenous community engagement and UNDRIP considerations. Problem statement workshop.
Fri	Space vs Non-Space Trade & Concept	Trade study methodology (criteria, weightings, scoring). Evaluating existing services (Radarsat, Sentinel, Planet) vs new missions. Constellation economics and business cases. Teams form, select a mission need, and run initial trade analysis in SpaceCDF.

WEEK 2: Concurrent Design Facility -- Designing a Mission

Day	Topic	Activities
Mon	System-V & Requirements	System-V model deep dive. SMART requirements from objectives. Functional decomposition. Teams enter mission need, run orbit/class advisors, generate requirements in SpaceCDF.
Tue	Subsystem Design: Power, AOCS,	Orbit selection trade. Solar array & battery sizing. Duty cycling. Attitude control selection. Thermal environment analysis. Teams run design loop, review parametric budgets.
Wed	Subsystem Design: Comms, Structure, Propulsion	Link budget calculation. Frequency band selection & licensing. CubeSat structure and deployer selection. Propulsion trade (if needed). Equipment selection in browser.
Thu	Integration, Verification, Cost	Interface matrix review. V&V matrix assignment. Environmental test planning. Cost estimation (parametric + COTS). BOM generation. Trade studies on key decisions.
Fri	Design Review & Optimization	Run optimizer. Sensitivity analysis. Gate review (MCR criteria). Resolve conflicts. Generate ECSS documents (MRD, TS, VP). Regulatory filings (ITU, RSSSA if applicable). Each team presents preliminary design.

WEEK 3: Mission Operations & Simulation

Day	Topic	Activities
Mon	Operations Concept	Ground segment architecture. MCS setup (COSMOS/OpenMCT/Yamcs overview). Operations procedures: nominal, contingency, FDIR. Pass planning and scheduling. Telecommand/telemetry definition from SpaceCDF FSW export.
Tue	Mission Operations Training	Console positions and responsibilities. Communication protocols (voice loops, log keeping). Anomaly response procedures. Dry-run walkthroughs of LEOP, nominal, and contingency scenarios. Practice commanding and telemetry monitoring.
Wed	Mission Simulation Day 1	Full-day simulated LEOP + commissioning. Teams operate their designed mission on console. Deployment, first contact, detumbling, initial checkout. Real-time anomaly injection by facilitators. Post-sim debrief and lessons learned.
Thu	Mission Simulation Day 2	Full-day simulated nominal operations + contingency. Science data acquisition mode. Ground pass downlink operations. Injected anomalies: safe mode entry, power anomaly, comms degradation. Constellation ops (for teams with multi-satellite designs). Post-sim debrief.
Fri	Wrap-Up & Presentations	Final design documentation review. Each team presents complete mission: need, design, ops concept, lessons learned. Peer review and Q&A. Certificate ceremony. Course evaluation. Discussion: next steps, collaboration opportunities, CSA/industry pathways.

Assessment

Component	Weight	Description
Worksheets (daily)	20%	Completed during sessions, formative feedback
Design Review (Week 2 Fri)	30%	Team presentation of preliminary design
Simulation Performance (Week 3)	20%	Console operations, anomaly response
Final Presentation (Week 3 Fri)	30%	Complete mission design, ops concept, lessons

Prerequisites

- Engineering background (any discipline) or equivalent experience
- Basic familiarity with orbital mechanics concepts (recommended, not required)
- Laptop with modern web browser (Chrome/Firefox/Edge)

Materials Provided

- SpaceCDF tool access (web-based, hosted for the programme)
- Facilitator's Book (reference, post-programme access)
- Learner's Workbook (worksheets, tool guides)
- ECSS/NASA standard excerpts (relevant sections)
- Canadian regulatory filing templates

Outcomes

Participants will leave with:

- A complete CubeSat mission design (documented, reviewed, ECSS-compliant)
- Hands-on experience with concurrent design methodology
- Understanding of Canadian space regulatory environment
- Mission operations experience including anomaly response

- Network of peers across industry and academia